SIMATS ENGINEERING
ASSESSMENT TOOL 1
COURSE NAME: Artificial Intelligence COURSE CODE: CSA1741

CO1-AT1-Analytical Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks: 50**

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

i. What are the percept's for this agent?
ii. Characterize the operating environment.
iii. What are the actions the agent can take?
iv. How can one evaluate the performance of the agent?
v. What sort of agent architecture do you think is most suitable for this agent? Why?

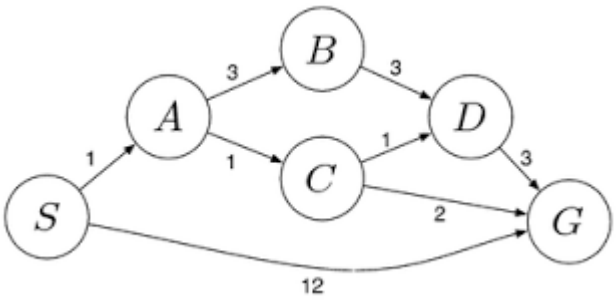
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path

from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I Niranjna devi S I / 192519100 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Niranjna devi S I

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

ANSWERS

1.Water-Jug Problem:

Let each state be given by (x,y). Here, (x) is the amount of water in the 4-gallon jug and (y) is the amount of water in the 3-gallon jug.

Water Jug Problem – Solution Table		
Step	**Action**	**State (4-gallon jug, 3-gallon jug)**
1	Initially, both jugs are empty.	(0, 0)
2	Fill the 3-gallon jug.	(0, 3)
3	Pour water into the 4-gallon jug.	(3, 0)
4	Fill the 3-gallon jug again.	(3, 3)
5	Pour water into the 4-gallon jug until it is full.	(4, 2)
6	Empty the 4-gallon jug onto the ground.	(0, 2)
7	**Pour the remaining water into the 4-gallon jug.**	**(2, 0)**

Using the above operations we have exactly 2 gallons of water in the 4-gallon jug. This problem shows how state space search can be used to reach a desired goal state by performing actions on an initial state.

2.MARS ROVER AGENT:

Mars Rover agent is an intelligent robotic agent used to explore Mars, take samples, analyse rocks and soils and send data back to earth.

i) What agent percept?

Percept are what the agent receives from its sensors:

Images of the Mars surface.
Navigation sensors to determine distance and obstacles.
Scientific equipment to analyse rock and soil make-up.
Temperature, air pressure and radiation levels.

Battery level, wheel status and system status.
Location, movement and communication.

ii) What is the environment?
The Mars Rover agent operates in a:

- Observability Partially observable - It cannot perceive everything.
- Deterministic Stochastic - Uncertainty in how the terrain and equipment will behave.
- Episodic / Sequential - Each action could affect future actions.
- Static / Dynamic - The environment can change.
- Discrete / Continuous Hybrid - Decisions may be discrete but movement and sensing is continuous.
- Single / Multi-agent Single (mostly) - It may interact with earth based agents.

iii) What actions can the agent perform?
The Mars Rover agent can:

- Move forward/backwards/turn.
- Take pictures of rocks and surrounding area.
- Take samples of rocks and soils.
- Analyse samples using scientific equipment.
- Avoid obstacles and choose safe routes.
- Send data and pictures back to earth.
- Save energy by recharging.
- Stop, wait or go into safe mode if faults are detected.

iv) How do we measure the agent's performance?

The performance can be measured by:

- How many useful samples were taken and analysed.
- Accuracy of data and how useful it is scientifically.
- Distance travelled.
- How many missions were successfully completed.
- Energy used and how efficiently it was used.
- Safety, reliability and no damage.
- Data sent back to earth.
- Performance Measure: The rover should aim to maximise scientific discoveries and complete missions whilst minimising energy use, risk and damage to equipment.

v) Which agent design would be best suited and why?

A model based, goal based and utility based agent.

Reasons:

- Model based - Has an internal model of the environment.
- Goal based - Plans actions to reach goals such as rocks/samples.
- Utility based - Compares route and actions based on safety, scientific value, energy use and time.
- Learning - Can learn from past observations to improve navigation and sample choices.
- Overall a hybrid intelligent agent would be suitable for exploring Mars autonomously.

3. Eight-Queens Problem – Search Space and Problem Formulation:

Problem: Place eight queens on an \(8 \times 8\) chessboard so that no two queens attack one another.

Queens must not share the same row, column, or diagonal.

Problem components:

Component	Description
Initial state	An empty chessboard with no queens placed.
State	A partial arrangement of queens on the chessboard.
Initial state	No queens have been placed.
Actions	Place a queen in a safe square in the next column.
Transition model	Adds a queen to the selected square.
Goal test	Eight queens are placed with no two attacking each other.
Path cost	Usually one unit per queen placement; all complete solutions have cost 8.

Search space exploration

A useful approach is backtracking search, a depth-first search strategy that places one queen per column.

1. Start with the first column.
2. Place a queen in a safe row.
3. Move to the next column and place another queen.
4. If no safe square exists, return to the previous column and move its queen to another row.
5. Repeat until all eight queens are placed safely.

For a state represented by [r_1,r_2,ldots,r_k], (r_i) denotes the row of the queen in column (i).

Two queens in columns (i) and (j) conflict diagonally when
[|r_i-r_j|=|i-j|]
They also cannot share a row, so (r_i ne r_j).

Example of a valid solution

One valid arrangement is:
[1,5,8,6,3,7,2,4]]
This means the queens are placed at:

- Column 1, row 1
- Column 2, row 5
- Column 3, row 8
- Column 4, row 6
- Column 5, row 3
- Column 6, row 7
- Column 7, row 2
- Column 8, row 4

This arrangement satisfies the row, column, and diagonal constraints.

Search strategies

- Breadth-First Search (BFS): Explores states level by level but can require considerable memory.
- Depth-First Search (DFS): Explores one branch at a time.
- Backtracking: Uses DFS and rejects unsafe partial arrangements immediately.
- Heuristic search: Can prioritize promising positions, although ordinary backtracking is sufficient for this problem.

  Conclusion: Backtracking is an effective method because it reduces the search space by eliminating invalid arrangements as soon as conflicts arise.

4. OLA Cab Booking – Problem-Solving Agent:

Problem: A customer wants to travel from a source location to a destination using the OLA Cab booking application. The customer can select different cab types, such as Mini, Micro, Sedan, Shared, or Prime.

Type of problem-solving agent:

A goal-based problem-solving agent with cost-based search and utility-based decision-making is suitable.

- Goal: Reach the destination successfully.
- Initial state: Customer's current location.
- Actions: Select cab type, choose a route, book a cab, and travel.
- Constraints: Fare, availability, travel time, comfort, and customer preference.
- Goal test: The customer reaches the destination.
- Performance measure: Minimize cost and travel time while satisfying the customer's comfort and safety requirements.

```
BEGIN

   INPUT source_location
   INPUT destination_location
   INPUT customer_preference
   INPUT maximum_budget

   IF source_location = destination_location THEN
      DISPLAY "Already at destination"
      STOP
   END IF

   available_cabs = GET_AVAILABLE_CABS(source_location)

   FOR each cab IN available_cabs DO
      cab.fare = ESTIMATE_FARE(cab, source_location,
                        destination_location)
      cab.time = ESTIMATE_TRAVEL_TIME(source_location,
                              destination_location)
   END FOR

   suitable_cabs = FILTER available_cabs
               WHERE fare <= maximum_budget
               AND cab satisfies customer_preference

   IF suitable_cabs IS EMPTY THEN
      DISPLAY "No suitable cab available"
      STOP
   END IF

   selected_cab = SELECT_BEST_CAB(suitable_cabs,
                           fare, time, comfort)

   DISPLAY selected_cab details
   ASK customer to confirm booking

   IF customer confirms THEN
      booking = BOOK_CAB(selected_cab)

      IF booking is successful THEN
         TRACK_CAB_AND_ROUTE()
         DISPLAY "Cab booked successfully"
      ELSE
         DISPLAY "Booking failed. Try another cab."
```

```
    END IF
  ELSE
    DISPLAY "Booking cancelled"
  END IF

END
```

Conclusion: The agent evaluates available cabs according to cost, time, and customer preferences, then selects a suitable option to achieve the travel goal.

5. Uniform Cost Search (UCS) – Solution:

From the given graph, the directed edges and their costs are:

Route	**Cost**
S → A	1
S → G	12
A → B	3
A → C	1
B → D	3
C → D	1
C → G	2
D → G	3

Step-by-step UCS execution:

UCS always expands the node with the lowest cumulative path cost.

Step	**Expanded node**	**Frontier (node, cumulative cost)**
1	S (0)	A(1), G(12)
2	A (1)	C(2), B(4), G(12)
3	C (2)	D(3), G(4), B(4)
4	D (3)	G(4), B(4), G(6)
5	G (4)	Goal reached

Calculate the possible path costs

1. $S \to G = 12$

2. $S \to A \to B \to D \to G$

Total cost $= 1 + 3 + 3 + 3 = \mathbf{10}$

3. $S \to A \to C \to D \to G$

Total cost $= 1 + 1 + 1 + 3 = \mathbf{6}$

4. $S \to A \to C \to G$

Total cost $= 1 + 1 + 2 = \mathbf{4}$

Final answer

Optimal path found using UCS

S → A → C → G

Optimal path found using UCS
S → A → C → G
Minimum total cost:(1+1+2= 4)

Conclusion: Uniform Cost Search finds the least-cost path from warehouse S to warehouse G as S → A → C → G, with a minimum total delivery cost of 4 units.